A Fluctuations-Based Approach for Extracting the Speed of Sound from Heavy Ion Collisions using *Trajectum* – a Hydrodynamical Model

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Introduction

Relativistic hydrodynamic simulations, like *Trajectum*[1], are used to describe heavy-ion collision observables. We replicate the conditions of the Relativistic Heavy Ion Collider (RHIC) and the Large Hadron Collider (LHC) to study the thermodynamic properties of the quark-gluon plasma (QGP) at $\sqrt{s_{NN}}$ = 5.02 TeV, 200 GeV, and 54.4 GeV. We do this by comparing the results of two distinct methods to clarify the conditions in which the thermodynamics can be extracted: one based on event-class-averaged observables, and a second based on event-by-event fluctuations.

Theory

The speed of sound as a thermodynamic state variable[2,3]:

$$c_s^2 = \frac{dP}{d\epsilon} = \frac{d\ln T}{d\ln s}$$

$$c_s^2 = \frac{d\ln\langle p_T\rangle}{d\ln\langle N_{ch}\rangle} \Rightarrow \langle p_T\rangle \propto \langle N_{ch}\rangle^{c_s^2} \qquad \frac{\Delta p_T}{\langle p_T\rangle_0} = c_s^2 \frac{\Delta N_{ch} + \delta}{\langle N_{ch}\rangle_0}$$

(event-class averages) *(event-by-event observables)*

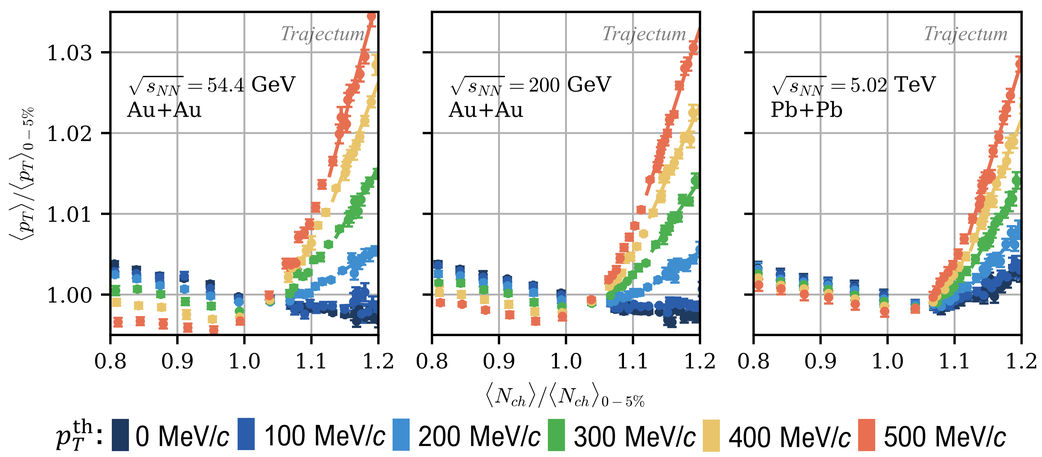
Mean transverse momentum $\langle p_T\rangle$ and charged-particle multiplicity $\langle N_{ch}\rangle$, as well as the fluctuations Δp_T and ΔN_{ch} of these observables, can be used as proxies for thermodynamic quantities. In the *ultra-central collisions*, collision geometry (volume) is **fixed → intrinsic fluctuations δ are sensitive to the QGP thermodynamics.**

Traditionally, c_s^2 has been equated with the slope of the $\langle p_T\rangle$-$\langle N_{ch}\rangle$ correlation. However, this slope **depends strongly on the centrality estimator (CE) definition** – particularly on the **threshold momentum** p_T^{th} on the CE, e.g.: $N_{ch} \in p_T^{\text{th}} < p_T < 10$ GeV/*c*, $|\eta| < 0.5$.

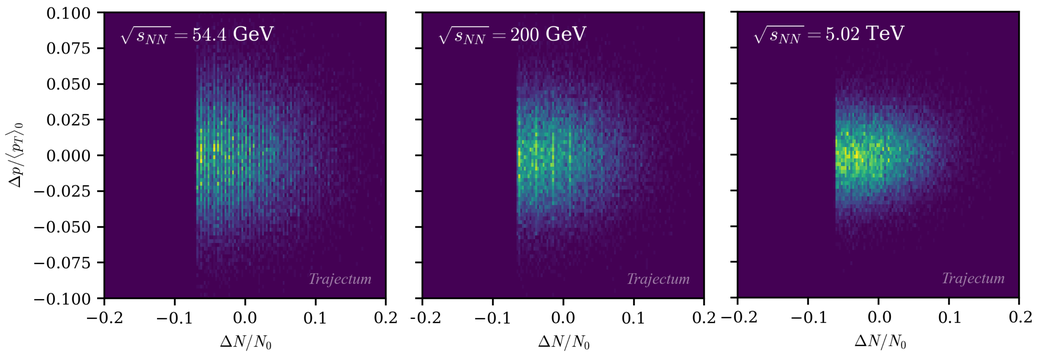
Analysis

Solution: compare the **CE dependent**, event-class averaged methodology to a **CE independent**, event-by-event methodology.

Power-law method *(CE-dependent)*

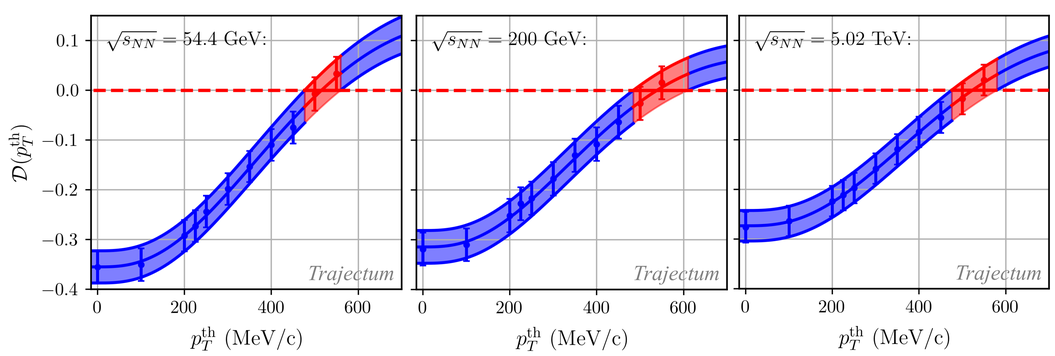


Event-by-event method *(CE-independent)*



Above: The joint distribution of normalized Δp_T and ΔN_{ch} in the 0-2.5% event class at $\sqrt{s_{NN}}$ = 5.02 TeV, 200, and 54.4 GeV.

Agreement criterion: $\mathcal{D}(p_T^{\text{th}}) \equiv c_{s\,PL}^2 - c_{s\,EBE}^2 = 0$



Discussion

Result: the choice of p_T^{th} in which the two methods are consistent, i.e.: in which an ultra-central collision measurement is sensitive to the thermodynamics of the QGP, is **finite.**

Physics interpretation:

- **Lower bound:** highly-inclusive CEs fail to resolve the fixed volume of ultra-central collisions[4] → sensitivity to c_s^2 is lost.
- **Upper bound:** high-p_T particles generally do not carry information about the equilibrium hydrodynamic phase[5] → sensitivity to c_s^2 is lost.

Limitations:

- Are hydrodynamical models like *Trajectum* applicable to all collision energies?

References

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